

- IPv4 Addresses
- Basic Subnetting
- **Mac Addresses**
- IPv6 Addresses

The MAC address

- A **media access control address (MAC address)** of a device is a unique identifier assigned to network interfaces for communications at the data link layer of a network segment.
- MAC addresses are also known as **hardware** or **physical addresses**.
- They uniquely identify an *adapter on a LAN* thus allow computers to uniquely identify themselves on a network.
- In computer networking, the MAC address is every bit as important as an IP address.
- MAC addresses function in the *media access control protocol sublayer* of the data link layer (layer 2) in the OSI model).

- A MAC address is made up of **12-digit hexadecimal numbers** which are 48 bits in length. Thus made up of 6 *octets* of 8 bits each.
- By convention, MAC addresses are usually written in one of the following formats:

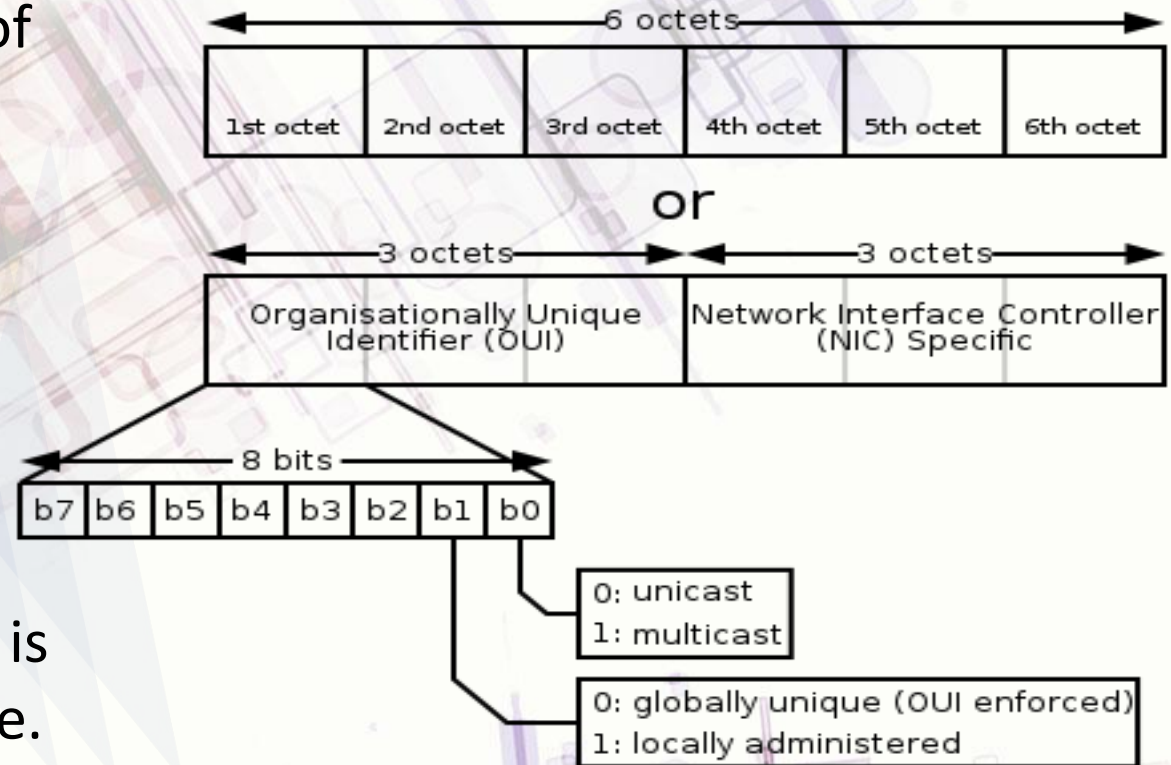
MM:MM:MM:SS:SS:SS

MM-MM-MM-SS-SS-SS

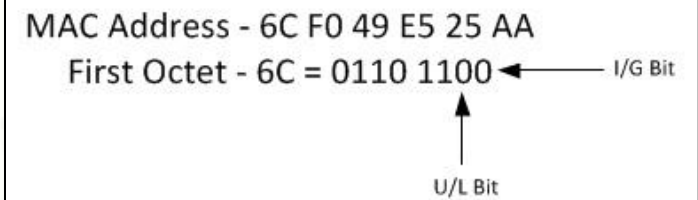
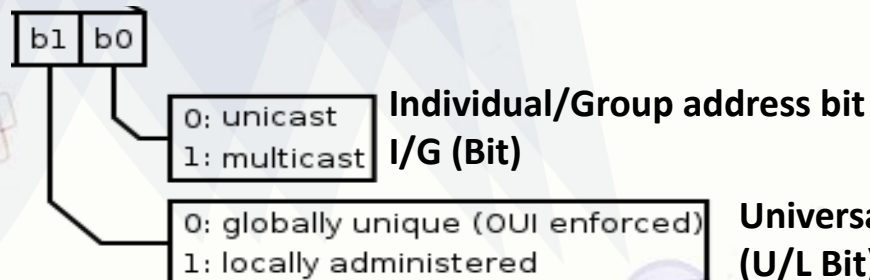
MMM.MMM.SSS.SSS

- A network node may have multiple NICs and each NIC must have a unique MAC address.
- Sophisticated network equipment such as a *multilayer switch* or *router* may require one or more permanently assigned MAC addresses.

- A MAC address has two major components:
- The *first half* of 24 bits and identifies the *manufacturer of the hardware. i.e. the Organizational Unique Identifier (OUI)*
- E.g. in **00:A0:C9:14:C8:29**, the prefix 00A0C9 indicates the manufacturer as Intel Corporation.
- The *second half* of 24 bits identifies the *device and represents the serial number assigned to the adapter by the manufacturer.*
- The requirement is for it to be unique.



- There are two bits in the first octet of a MAC address that are used to define certain aspects of the MAC address.
- The **least significant bit** (8th bit) of the first octet of the MAC address is the **Individual/Group(I/G)** address bit. It is used to signify if the destination MAC address is a *unicast* or a *multicast* address.
 - If bit is 0 - *Individual MAC address* and is a *unicast address*.
 - If bit is 1 - *Group address* and is a *multicast address*.
- The **second least significant bit** (7th bit) of the first octet of the MAC address is the **Universally or Locally administered address bit (U/L) bit**. It is used to tell if the MAC address is the *burned-in-address(BIA)* or a MAC address that has been changed locally.
 - If bit is 1 - MAC address has been changed -*locally set up/ administered*.
 - If bit is 0 -MAC address is universally administered/globally unique (BIA).



- MAC addresses are formed according to the rules of one of three numbering name spaces managed by the Institute of Electrical and Electronics Engineers (IEEE): **MAC-48**, **EUI-48**, and **EUI-64**.
- The IEEE claims trademarks on the names EUI-48 and EUI-64, in which **EUI** is an abbreviation for **Extended Unique Identifier**.
- The distinction between EUI-48 and MAC-48 identifiers is purely nominal:
 - ✓ *MAC-48 is used for network hardware;*
 - ✓ *EUI-48 is used to identify other devices and software.*
- The IEEE now considers the label MAC-48 to be an obsolete term.



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- **IPv6 Addresses**

Brief History

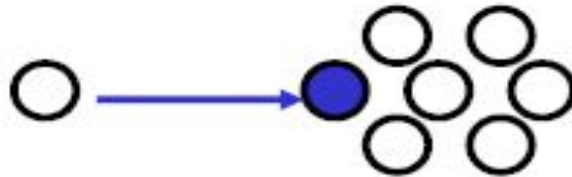
- With admirable foresight, the Internet Engineering Task Force (IETF) started the Internet Protocol Next Generation (or *IPng*) area in 1993 to investigate the different proposals and to make recommendations for further procedures.
- As early as in 1994, they initiated the design and development of a suite of protocols and standards now known as Internet Protocol Version 6 (IPv6), as a worthy tool to phase out and supplant IPv4 over the coming years.
- There is an explosion of sorts in the number and range of IP capable devices that are being released in the market and the usage of these by an increasingly tech savvy global population.
- IPv6, the new protocol was developed to effectively support the ever-expanding Internet usage and functionality, and also address security concerns.
- There will be no “flag day” for IPv6 like there was for the 1983 move from NCP to IPv4. IPv6 will slowly and gradually grow into our networks and the Internet. Taking a step-by-step approach to IPv6 may be the most cost-efficient and risk-averse way to integrate it.

IPv6 Addressing Modes

- IPv6 designers carefully created IPv6 addressing modes to support facilities that *ease network management* by enabling nodes to *configure automatically*, to *reduce the effort required to renumber a network*, and to *increase the efficiency of Internet backbone routers*.
- Thus IPv6 goes way beyond simply expanding the address space.
- Recognizing that the transition from IPv4 to IPv6 will take many years, the designers provided formats to enable IPv4 and IPv6 to coexist.
- There are **three modes of IPv6 addressing** - *unicast, multicast and anycast*, each of which can be classified by the part of the network over which it is valid.
- In addition, there is a **fourth addressing mode** *IPv4 mapped*, which is used in the transition from IPv4 to IPv6. As in the case of IPv4, a network interface may support multiple addresses at the same time. It may support addresses of each of the various modes and may support multiple addresses of the same mode.

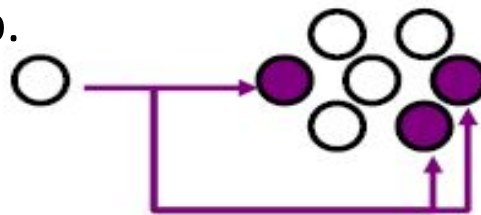
Unicast Addresses

- IPv6 Unicast addresses are identical to IPv4 unicast addresses. A packet sent to a unicast address will be received by only one network node.



Multicast Addresses

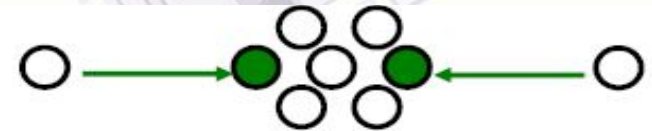
- Multicast addresses are used in the same way as multicast addresses in IPv4. Multicast was added to IPv4 after the basic address formats were designed, these were experimental, thus are confined to a very limited portion of the IPv4 address space.
- In IPv6 they are integral, and will provide ample space for multicast addresses. The packet sent to a multicast address is delivered to members of a group.



NB: There are no broadcast addresses in IPv6. Multicast replaces broadcast.

Anycast Addresses

- Anycast addresses are new in IPv6. Packets are routed to only one of the member of the group, the nearest one. A message sent to an anycast address is sent to only one of the interfaces identified by the anycast address. For example, all of the DNS servers at a site may share a single anycast address since all provide identical service. A DNS request sent to that address will go to whichever server is closest to the requesting node in terms of the site's internal routing protocol.



IPv4 mapped:

- IPv4 mapped addresses are used to support the IPv4 to IPv6 transition. It is used to enable IPv6 devices to communicate through IPv4 connected networks.
- The format for IPv4 Mapped address has the first 80 bits set to zeros, followed by the next 16 bits set to all ones and finally, the last 32 bits written in dotted decimal appended to then end forming 128 bit IPv6 address as illustrated below.
- E.g. An IPv4 Class A address of 12.155.166.101 would look like this in IPv4 Mapped address **0000:0000:0000:0000:0000:FFFF:12.155.166.101** or **::FFFF:12.155.166.101** in IPv6's short form.

Unicast Address schemes

- IPv6 has three different types of Unicast Address schemes – *Global Unicast address, link-local address* and *unique local address*.
- We shall look at these but let's learn something interesting first

Extended Unique Identifier (EUI-64) Interface ID

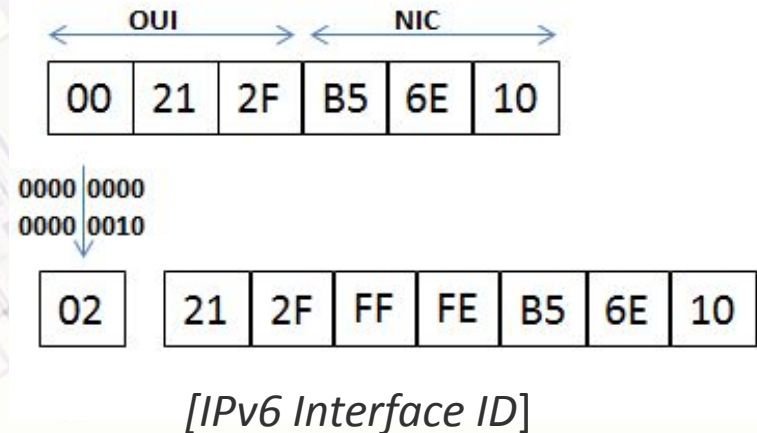
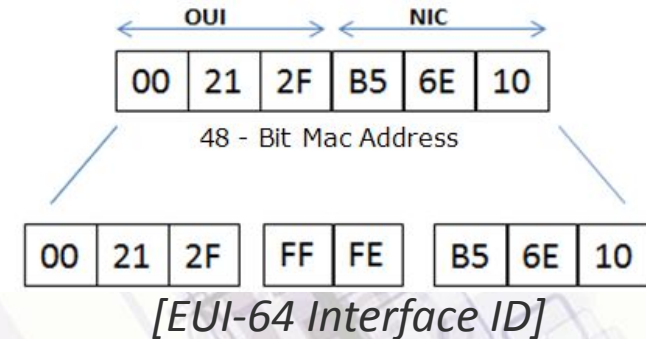
- The second half of an IPv6 unicast address (last 64 bits) is always used for *Interface ID*. MAC addresses are considered to be uniquely assigned worldwide. Interface ID takes advantage of this uniqueness of MAC addresses. A host can auto-configure its Interface ID by using IEEE's Extended Unique Identifier (EUI-64) format.

How?

- First, the host divides its own MAC address into two 24-bits halves.
- Then 16-bit Hex value 0xFFFE is sandwiched into those two halves of MAC address, resulting in *EUI-64 Interface ID*.
- This Extended Unique Identifier (EUI), allows a host to assign itself a *unique 64-Bit IP Version 6 interface identifier (EUI-64)*.

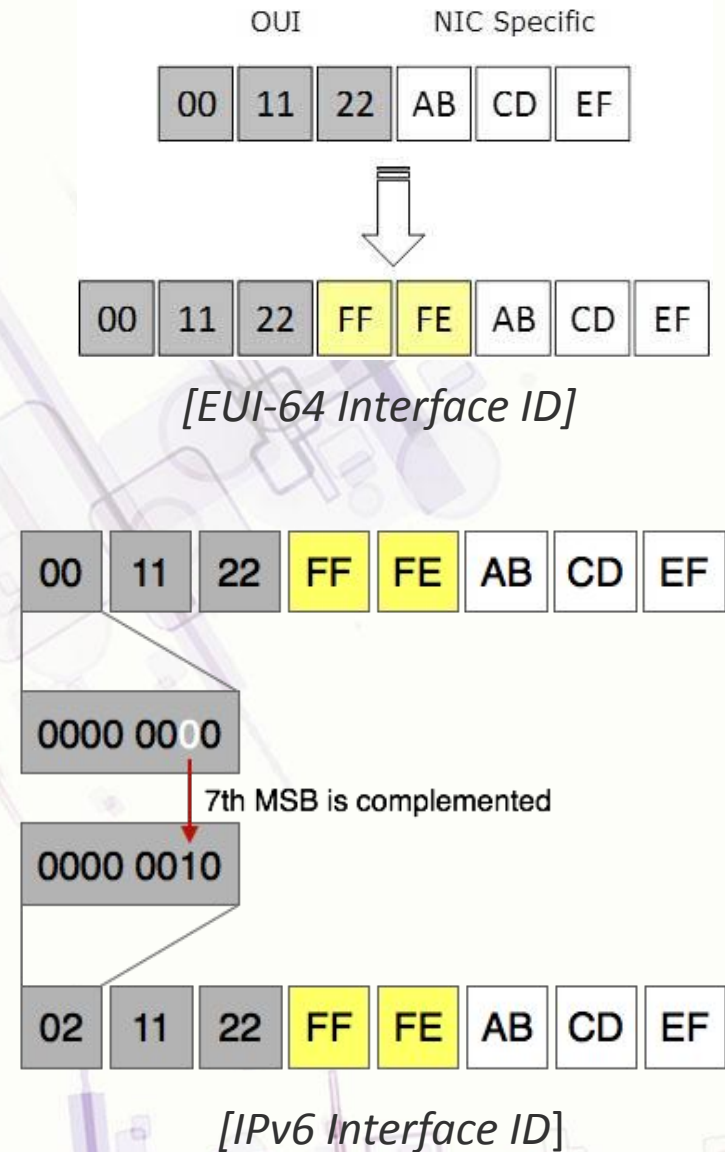
Let's break the process:

- First, the sequence FF-FE is inserted in between the OUI and the NIC identifier. IEEE has chosen FFFE as a reserved value which can only appear in EUI-64 generated from an EUI-48 MAC address.
- Next, the **Universal/Local bit (U/L bit)**, -7th bit, needs to be inverted. This bit identifies how the address is administered.
- If the bit is 0, the address is globally /universally administered. (default)
- If it is 1, the address is locally administered.
- Local administration of the 7th bit eliminates the need for an *address resolution protocol* since the MAC address for any host can be simply determined, hence no manual configuration/ DHCP as done in IPv4.



Example 2:

- First, the sequence FF-FE is inserted in between the OUI and the NIC identifier.
- Next, to convert EUI-64 ID into IPv6 Interface Identifier, the most significant 7th bit of EUI-64 ID is complemented. i.e. the **Universal/Local bit (U/L bit)**, -7th bit, needs to be inverted. This bit identifies how the address is administered.
 - If the bit is 0, the address is globally/ universally administered.
 - If it is 1, the address is locally administered.



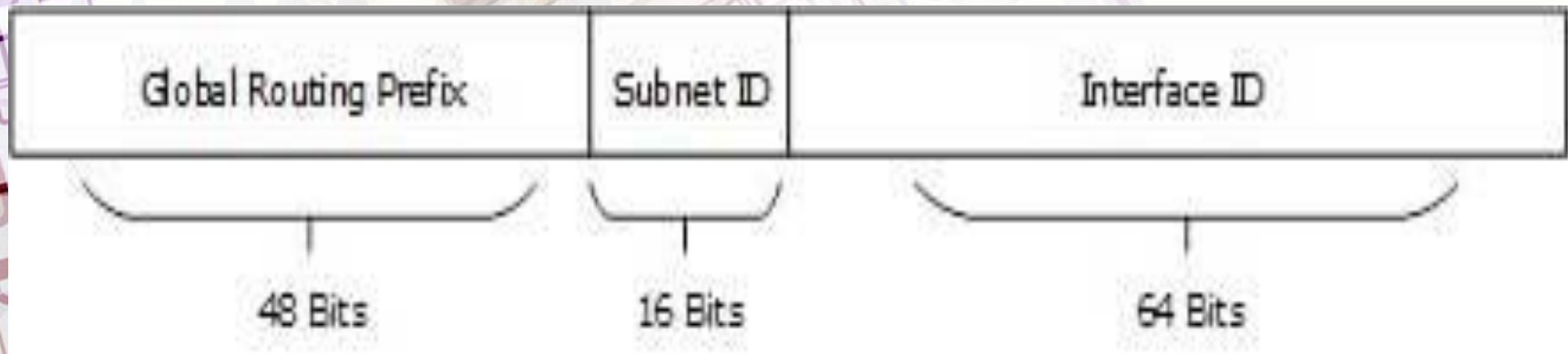
IPv6 Netmask and Hostmask

- IPv6 addresses are split in two parts
 - ✓ Network part (netbits)
 - ✓ Host part (hostbits)
- IPv4's netmask is like 255.255.255.0 format
- The **netmask** for an IPv6 address is in 2001:638:A01:2::/64 where 64 is the length of the netbits. The **netbits** are referred to as the "prefix"
- Generally, the structure of an IPv6 address is as follows in sequence -48 netbits, 16 subnetbits, and 64 hostbits

Unicast Address schemes

Global Unicast Address

- This address type is equivalent to IPv4's public address. Global Unicast addresses in IPv6 are globally identifiable and uniquely addressable.
- **Global Routing Prefix:** The most significant 48-bits are designated as Global Routing Prefix which is assigned by a provider to an organisation. The three most significant bits of Global Routing Prefix is always set to 001.



Unicast Address schemes

Link-Local Address

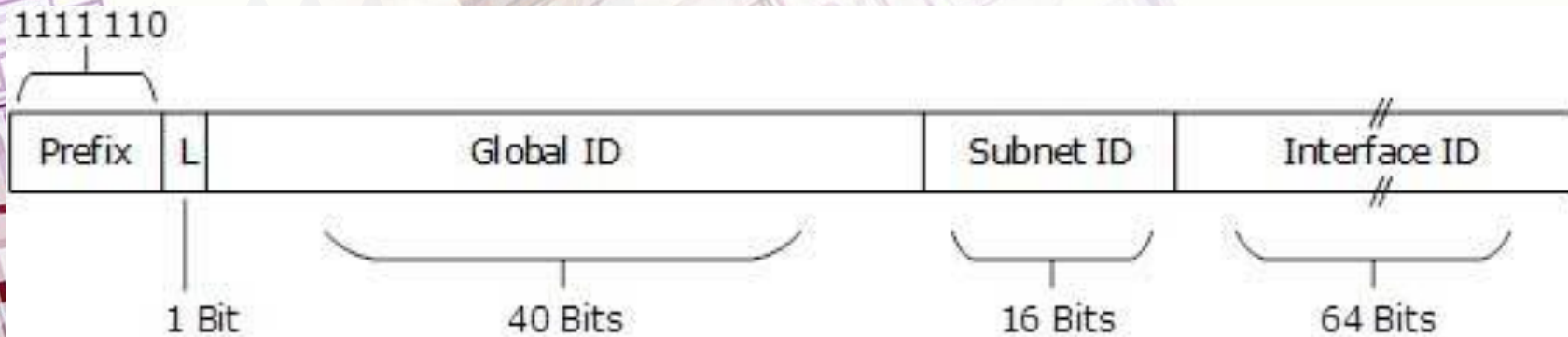
- Auto-configured IPv6 address is known as Link-Local address. This address always starts with FE80. The first 16 bits of link-local address is always set to 1111 1110 1000 0000 (FE80). The next 48-bits are set to 0, thus:
- Link-local addresses are used for communication among IPv6 hosts on a link (broadcast segment) only. These addresses are not routable, so a Router never forwards these addresses outside the link.

1111 1110 1000 0000 0000 0000 0000 0000 0000 0000 0000 0000 0000 0000 0000 0000	Interface ID
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Unicast Address schemes

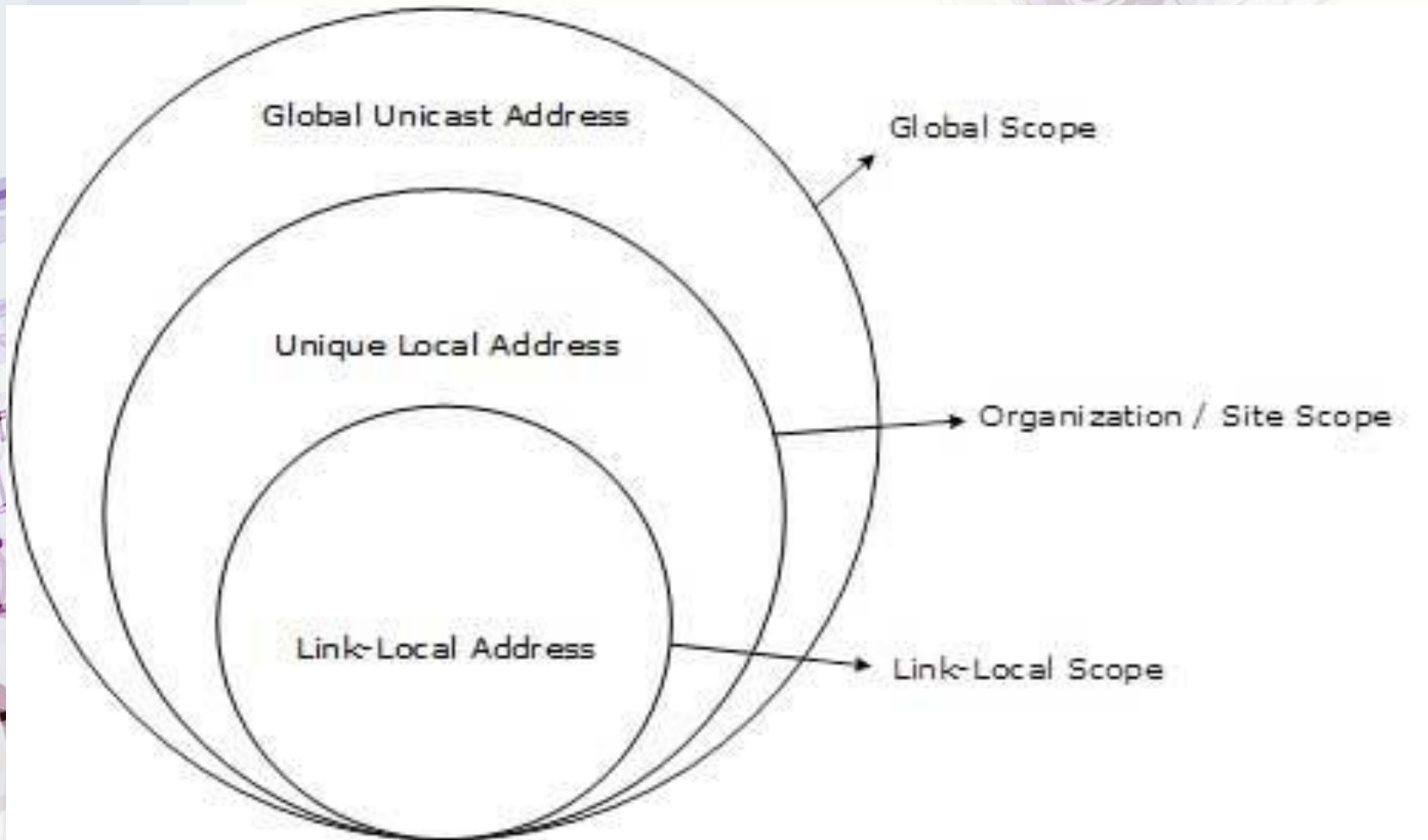
Unique-Local Address

- This type of IPv6 address is globally unique, but it should be used in local communication. The second half of this address contain Interface ID and the first half is divided among Prefix, Local Bit, Global ID and Subnet ID.
- Prefix is always set to 1111 110. L bit, is set to 1 if the address is locally assigned. So far, the meaning of L bit to 0 is not defined. Therefore, Unique Local IPv6 address always starts with 'FD'.



Unicast Address schemes

Scope of IPv6 Unicast Addresses:



Why a larger Address Space?*

- In February 2011, the last unassigned top-level address blocks of 16 million IPv4 addresses were allocated by the IANA to the five regional Internet registries (RIRs).
- As of September 2015, all of APNIC, RIPE_NCC, LACNIC, ARIN had reached this stage. AFRINIC is only RIR that is still using the normal protocol for distributing IPv4 addresses.
- IANA unallocated Address Pool Exhaustion: **03-Feb-2011**

- Projected RIR Address Pool Exhaustion Dates:

- RIR	Projected Exhaustion Date	Remaining Addresses
in RIR Pool (/8s)		
- APNIC:	19-Apr-2011 (actual)	0.2662
- RIPE NCC:	14-Sep-2012 (actual)	0.4495
- LACNIC:	10-Jun-2014 (actual)	0.1132
- ARIN:	24 Sep-2015 (actual)	
- AFRINIC:	12-May-2019	0.4514

Source: <https://en.wikipedia.org/wiki/IPv6>, <https://ipv4.potaroo.net/>

Why a larger Address Space?*

- As of December 2017, the *world population* was estimated to have reached 7.6 billion. The UN estimates it will reach 11.8 billion by 2100.
- As of June 30 2017, the total number of *world internet users* was 3,885,567,619, just over half of the world's population.
- IPv6 - (2^{128}) 340,282,366,920,938,463,463,374,607,431,768,211,456 addresses, 7.9×10^{28} times more as many as IPv4 - (2^{32}) 4,294,967,296.
- There is enough /48 IPv6 address prefixes available to give every person on the planet about 4000 allocations before IANA has to release some more of the 80% of the space which is still undefined!
- In 2015, there were about 15.4 billion connected devices this number will be 30-50 billion in 2020, 75.4 billion by 2025 & 125 billion in 2030.
- In 2016 about 29 million units of wearable devices were sold.
- This connected lifestyle means that there could be 500 devices in each home connected to the web by 2022, with about 328 million devices being connected to the internet each month.

IPv6 vs IPv4 (*Why IPv6?*)

- *Exhausted IPv4 address space:* As of 2011, IPv4 address space was virtually exhausted despite NAT and CIDR. Thus *increase the IP addressing space* from 32 bits to 128 bits . Each user/person can have up to 5×10^{28} addresses.
- *IPv4 addresses are non -hierarchical :* IPv4 addresses are non -hierarchical and assigned irrespective of geographical topology. This leads to fragmentation and thus big routing tables (as of 2010 over 320k route prefixes to be exchanged between backbone routers). IPv6 provides new functions at the IP layer: source routing, QoS, auto-configuration, security, etc.
- *Disproportionate IPv4 address assignment :* IPv4 addresses are assigned disproportionately (2005: USA 75%, Asia only ~10%, China < 1%).
- *IPv4 address management is difficult:* IPv4 does not really support automatic address assignment (except APIPA). Usage of DHCP means high administrative effort. IPv6 provides *permanent IP addresses*, instead of using DHCP or NAT.

IPv6: Key Features

- **Larger Address Space:** In contrast to IPv4, IPv6 uses 4 times more bits to address a device on the Internet. This much of extra bits can provide approximately 3.4×10^{38} different combinations of addresses. This address can accumulate the aggressive requirement of address allotment for almost everything in this world. According to an estimate, 1564 addresses can be allocated to every square meter of this earth.
- **Enhanced Priority Support:** IPv4 used 6 bits DSCP (Differential Service Code Point) and 2 bits ECN (Explicit Congestion Notification) to provide Quality of Service but it could only be used if the end-to-end devices support it, that is, the source and destination device and underlying network must support it. In IPv6, Traffic class and Flow label are used to tell the underlying routers how to efficiently process the packet and route it.
- **Mobility:** IPv6 was designed keeping mobility in mind. This feature enables hosts (such as mobile phone) to roam around in different geographical area and remain connected with the same IP address. The mobility feature of IPv6 takes advantage of auto IP configuration and Extension headers.

IPv6: Key Features

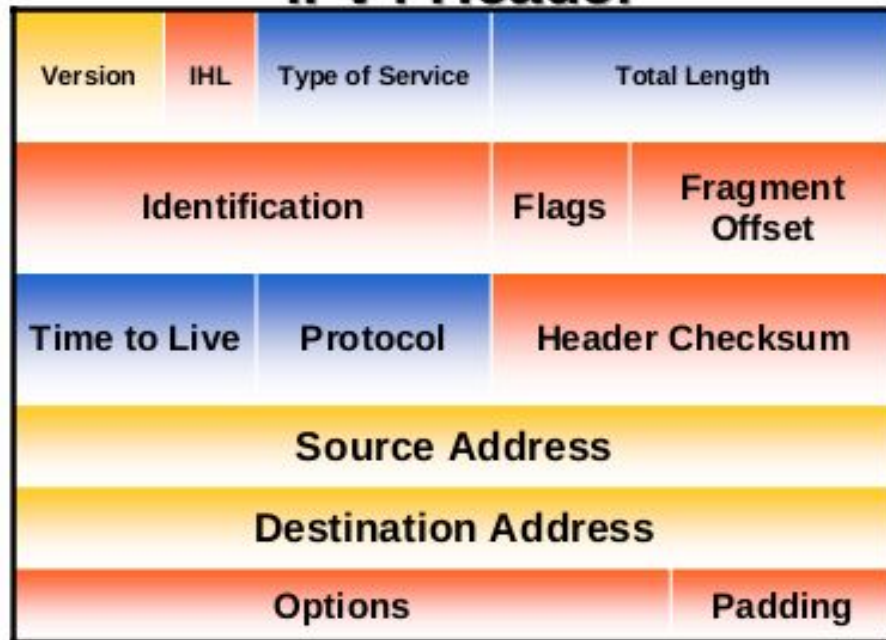
- *Auto-configuration*: IPv6 supports both stateful and stateless auto configuration mode of its host devices. This way, absence of a DHCP server does not put a halt on inter segment communication.
- *Faster Forwarding/Routing*: Simplified header puts all unnecessary information at the end of the header. The information contained in the first part of the header is adequate for a Router to take routing decisions, thus making routing decision as quickly as looking at the mandatory header.
- *No Broadcast*: Though Ethernet/Token Ring are considered as broadcast networks because they support Broadcasting, IPv6 does not have any broadcast support any more. It uses multicast to communicate with multiple hosts.
- *Anycast Support*: This is another characteristic of IPv6. IPv6 has introduced Anycast mode of packet routing. In this mode, multiple interfaces over the Internet are assigned same Anycast IP address. Routers, while routing, send the packet to the nearest destination.
- *IPSec*: Initially it was decided that IPv6 must have IPSec security, making it more secure than IPv4. This feature has now been made optional.

IPv6: Key Features

- *Extensibility*: One of the major advantages of IPv6 header is that it is extensible to add more information in the option part. IPv4 provides only 40-bytes for options, whereas options in IPv6 can be as much as the size of IPv6 packet itself.
- *End-to-end Connectivity*: Every system now has unique IP address and can traverse through the Internet without using NAT or other translating components. After IPv6 is fully implemented, every host can directly reach other hosts on the Internet, with some limitations involved like Firewall, organization policies, etc.
- *Simplified Header*: IPv6's header has been simplified by moving all unnecessary information and options (which are present in IPv4 header) to the end of the IPv6 header. IPv6 header is only twice as bigger than IPv4 provided the fact that IPv6 address is four times longer. This means routers can take forwarding decisions and forward them as quickly as they arrive.
- *Smooth Transition*: Large IP address scheme in IPv6 enables to allocate devices with globally unique IP addresses. This mechanism saves IP addresses and NAT is not required. So devices can send/receive data among each other, e.g. VoIP, streaming much efficiently.

Header comparison: IPv4 vs IPv6

IPv4 Header



IPv6 Header



- Legend**
-  - Field's name kept from IPv4 to IPv6
 -  - Fields not kept in IPv6
 -  - Name & position changed in IPv6
 -  - New field in IPv6

Abbreviation

- For convenience, an IPv6 address may be abbreviated to shorter notations by application of the following rules.
- ✓ One or more leading zeroes from any groups of hexadecimal digits are removed; this is usually done to either all or none of the leading zeroes. 16-byte IPv6 address written as 8 groups of 4 hexadecimal digits with colons between groups

8000:0000:0000:0000:0123:4567:0000:CDEF

- ✓ Consecutive sections of zeroes are replaced with a double colon (::). The double colon may only be used once in an address, as multiple use would render the address indeterminate. It is recommended that a double colon not be used to denote an omitted single section of zeroes.

0123 123

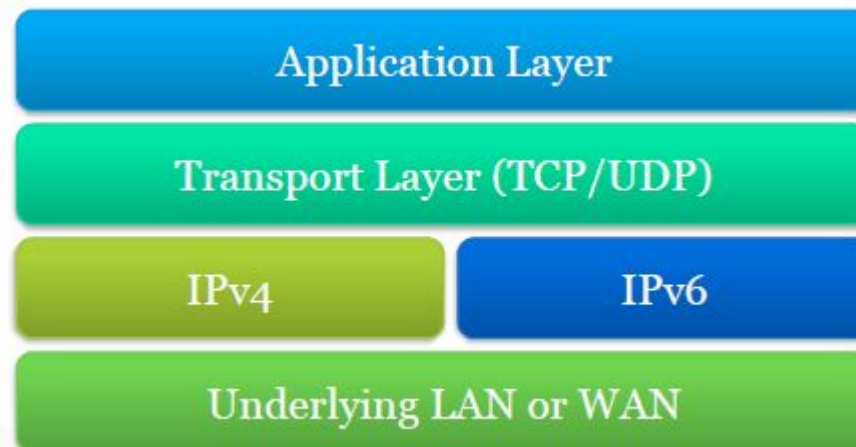
- One or more groups with 0s can be replaced by a pair of colons.

8000::123:4567:0:CDEF

- The loopback address below is abbreviated to ::1 using both rules
0000:0000:0000:0000:0000:0000:0000:0001 abbreviated to ::1

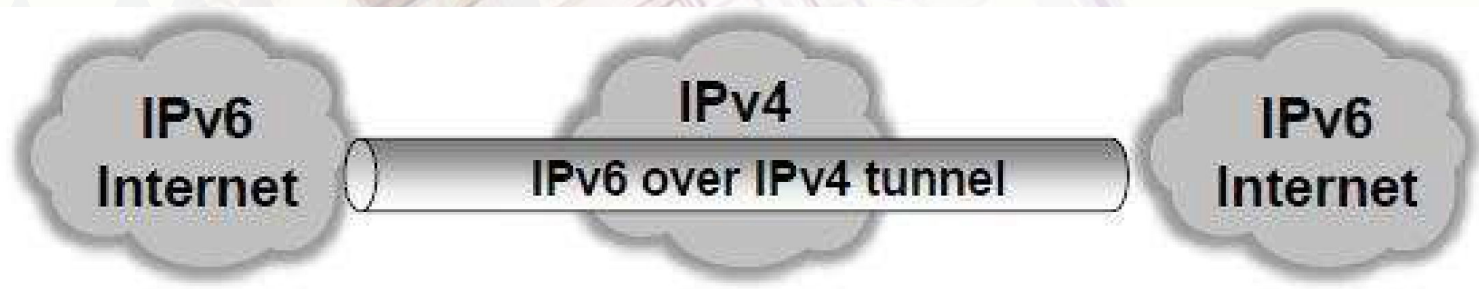
Transition Mechanisms

- Till now IPV6 is not globally accepted. So transition is needed from IPV4 to IPv6. Among the several strategies deployed are dual stack, tunneling and header transition.
- **Dual Stack**
 - All hosts must have a dual stack of protocols before migrating completely to version 6.
 - DNS determines which version to use when sending a packet.



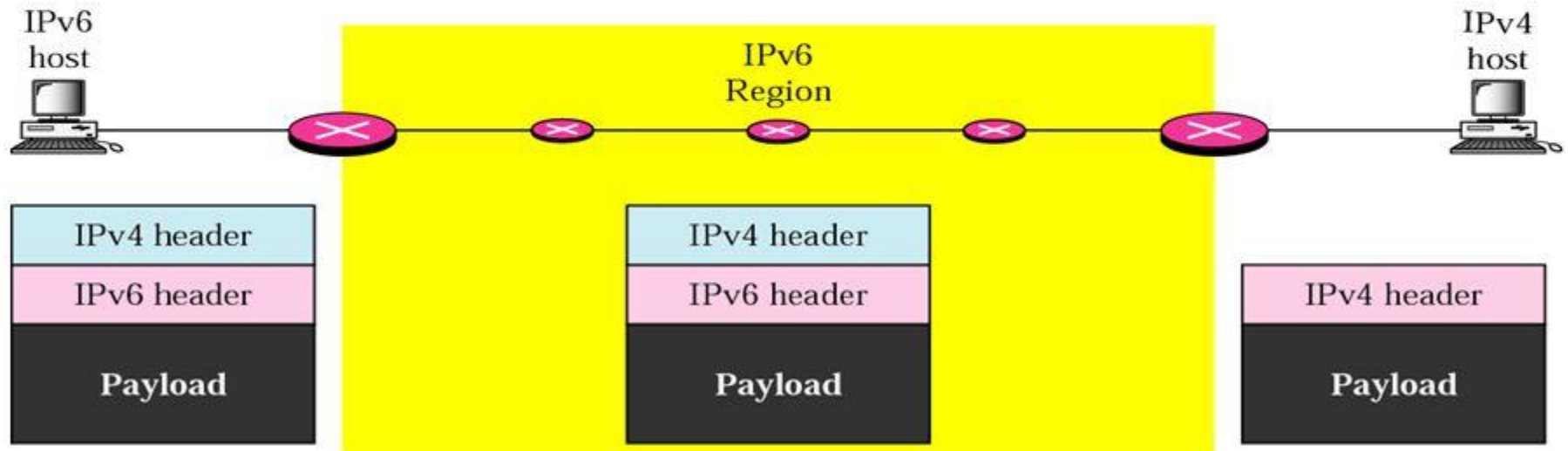
- **Tunnelling**

- Two computer using IPV6 want to communicate with each other and the packet must pass through an IPv4 region.
- The IPv6 packet is encapsulated in an IPv4 packet when it enters the region and leaves its capsule when its exits the region.



• Header Translation

- This is necessary for when majority of the Internet has moved to IPv6 but some systems still use IPv4.
- The header format must be converted to IPv4.
- It uses the mapped address to translate an IPv6 address to an IPv4 address.



~Quiz~

1. Abbreviate the following IPv6 Addresses
 - a) 2000:0000:0000:0000:1234:0000:0000:000B
 - b) ABCD:0000:0000:0000:0000:0000:0000:0173
 - c) FE80:0000:0000:0100:0000:0000:0000:0123
 - d) 2345:0123:4040:0000:0000:0000:000A:000B
2. Write the following 48-bit MAC addresses in the 3 possible conventional formats. For each, indicate the OUI and the device ID.
 - a) 01-23-ab-cd-ef-45
 - b) D8-D3-85-EB-12-E3
 - c) 12-34-56-78-9A-BC
 - d) 00:1B:44:11:3A:B7
3. Translate the following 48-bit networks to 64-bit networks
(Hint: by inserting an FFFE value in the middle. E.g. 00:0C:29:0C:47:D5 48-bit address converted to a 64-bit address will be 00:0C:29:FF:FE:0C:47:D5)
 - a) 01-23-ab-07-cd-ef
 - b) D8-D3-85-EB-12-E3
 - c) 12-34-56-78-9A-BC
 - d) 00:1B:44:11:3A:B7
4. Name and explain *four advantages* that IPv6 has over IPv4.
5. Name and describe any *six unique features* of IPv6.
6. Describe *three mechanisms* deployed to transition from IPv6 to IPv4. Include diagrams.

~Reading Assignment~

Read and make your own brief additional notes on the following.

NB: The topic is examinable hence the notes will form part of your reading material.

1. MAC addresses
 - a) EUI-48
 - b) MAC-48
 - c) EUI-64
2. IPv6 Addresses
 - a) Key features
 - b) Representation/Abbreviation
 - c) Types/Scopes (Unicast, Multicast, Anycast)
 - d) Address formats (for Unicast & Anycast-same, and for Multicast)
 - e) Transition Mechanisms (Tunnelling, Dual stacking, header translation)